

Python Study Plan (Job-Oriented, Hands-On)

Phase 1: Core Python

Topic	Focus / Outcome
Syntax & Basics	Variables, data types, operators
Control Flow	if / for / while
Functions	Arguments, return, scope
Data Structures	list, tuple, dict, set
map / filter / reduce	Functional constructs
File Handling	Read / write files
__main__ guard	Script vs module
Context Managers	with, contextlib

Phase 2: Pythonic Thinking

Topic	Why It Matters
Comprehensions	Cleaner & faster code
Generators & yield	Lazy execution, memory efficient
Iterators	Custom iteration
Decorators	Cross-cutting concerns
Exceptions	Robust error handling
Logging	Production-grade debugging

Phase 3: Small Projects

Project	Skills Practiced
Log Analyzer	Files, dicts, loops
Weather Fetcher	API, requests, JSON
CSV Processor	csv module
File Renamer	OS automation
Password Generator	random, security basics

Phase 4: Backend Python

Topic	Tools
-------	-------

Web Framework	FastAPI
API Design	REST, JSON
Validation	Pydantic
Authentication	JWT basics
ORM	SQLAlchemy
Testing	pytest

Phase 5: Advanced / Job Readiness

Topic	Focus
Async Python	async / await
Type Hints	typing module
Environment Mgmt	virtualenv, poetry
Packaging	pip, wheels
Interview Prep	Python DSA